

Chapter 6.2

Hydrogenation of Imino Groups

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1 Introduction and Background

The enantioselective reduction of prochiral C=N double bonds to obtain the corresponding chiral amines still represents a major challenge. Whereas many highly enantioselective chiral catalysts are known for the enantioselective hydrogenation of C=C and C=O bonds, only relatively few effective reduction methods are available for the corresponding C=N functions [1, 2, 3, 4, 5]. Imines and related C=N functions have some chemical peculiarities that make their stereoselective reduction more complex than that of ketones or olefins. Firstly, even though the preparation starting from the corresponding amine derivative and carbonyl compound is relatively simple, complete conversion is not always possible and formation of trimers or oligomers can occur. Then, the resulting C=N compounds are often sensitive to hydrolysis and the presence of enamine as well as *syn/anti* isomers can be a problem as demonstrated for the hydrogenation of imines [6] and of oximes [7]. In addition, many of the homogeneous catalysts can complex with both the starting material and the product imine and as a consequence, catalytic activity is often low.

In this contribution we first give a short description of the historical development of enantioselective C=N hydrogenations. Then, an overview on effective enantioselective catalysts for different types of C=N groups is presented, that is directed to the synthetic chemist involved in synthesis planning. The detailed discussion of the chemistry of selected asymmetric catalysts is meant for the catalyst specialist. Finally, the most useful methods are briefly assessed from a preparative as well as a technical point of view.

2 Historical Development

Chiral amines were always considered important targets for synthetic chemists and attempts to prepare such compounds enantioselectively date back quite early [8]. Table 1 gives selected milestones for the development of enantioselective catalysts for the reduction of C=N functions. At first, only heterogeneous hydrogenation catalysts such as Pt black [9], Pd/C [10], or Raney nickel [11,12] were applied. They were modified with chiral auxiliaries in the hope that some induction, i.e., transfer of chirality from the auxiliary to the reactant might occur. These efforts were undertaken on a purely empirical basis without any understanding of what might influence the desired selectivity. Only very few substrate types were studied and not surprisingly, optical yields were low and could not always be reproduced. The first reports on homogeneous Ru [13] and Rh [14] catalysts appeared in 1975, but useful enantioselectivities were reported by the Marko group only in 1984 [15]. Remarkable progress has been made in the 1990s and now, several very selective catalysts are available for different types of C=N functions. Moreover, the first industrial process was announced in 1996 [20].

Table 1. Selected milestones for the enantioselective hydrogenation of C=N functions

Year	Substrate	Catalyst	Chiral auxiliary ^a	ee (%)	Comment	Ref
1941	oxime	Pt black	menthoxyacetic acid	3	first reported experiment	9
1958	dioxime	Pd	silk fibroin	15	chiral support	10
1975	oxime	Ru complex	diop	15	homogeneous Ru catalyst	13
1975	imine	Rh complex	diop	22	homogeneous Rh catalyst	14
1984	C=N-Alk	Rh complex	bdpp	72	first useful ee	15
1989	C=N-Alk	Rh complex	bdpp _{sulf}	94	first very high ee	16
1990	C=N-Ar	Ir complex	bdpp	84	homogeneous Ir catalyst	17
1992	cyclic imine	Ti complex	ebthi	99	homogeneous Ti catalyst	6
1992	C=N-X	Rh complex	duphos	96	acyl hydrazone	18
1996	C=N-X	Ru complex	N [^] N	97	transfer hydrogenation	19
1996	MEA-imine	Ir complex	xyliphos	80	first industrial application	20

^afor structures of ligands see Fig. 1

3

Synthetically Useful Catalysts for Various C=N Functions

The nature of the substituent directly attached to the N-atom influences the properties (basicity, reduction potential, etc.) of the C=N function more than the nature of the substituents at the carbon atom. For example, it was found that the Ti-ebthi catalyst (Fig. 1) can hydrogenate only *N*-alkylimines but not *N*-arylimines [6]. Oximes and other C=N-X compounds show even a more pronounced variation in their reactivity. The following sections give a short summary of the results obtained for different classes of C=N groups. Only catalysts with synthetically useful selectivities or otherwise of interest were included in Tables 2, 3, 4, 5, and 6 (s/c: substrate/catalyst ratio, tof: turnover frequency at high conversion).

3.1

Enantioselective Reduction of *N*-Arylimines

Most efforts have been concentrated on 2,6-dialkyl-substituted-*N*-arylimines and several of these can be hydrogenated with good ees and modest to very high catalyst activity (Table 2). The reason for this interest is the fact that the corresponding amines with a stereogenic C-atom in an α -position to the nitrogen atom are intermediates for a number of important pesticides and that in many cases one stereoisomer carries most of the biological activity [21]. Because the corresponding ketimines are easily accessible, their enantioselective reduction has been investigated in much detail. In all cases, best results were obtained with Ir-diphosphine catalysts. With the exception of the Ir-bdpp catalyst (Table 2, entry 2.1) [22], the presence of extra iodide ions was usually necessary. For the Ir-ferrocenyldiphosphine catalysts [20], addition of acid led to a dramatic in-

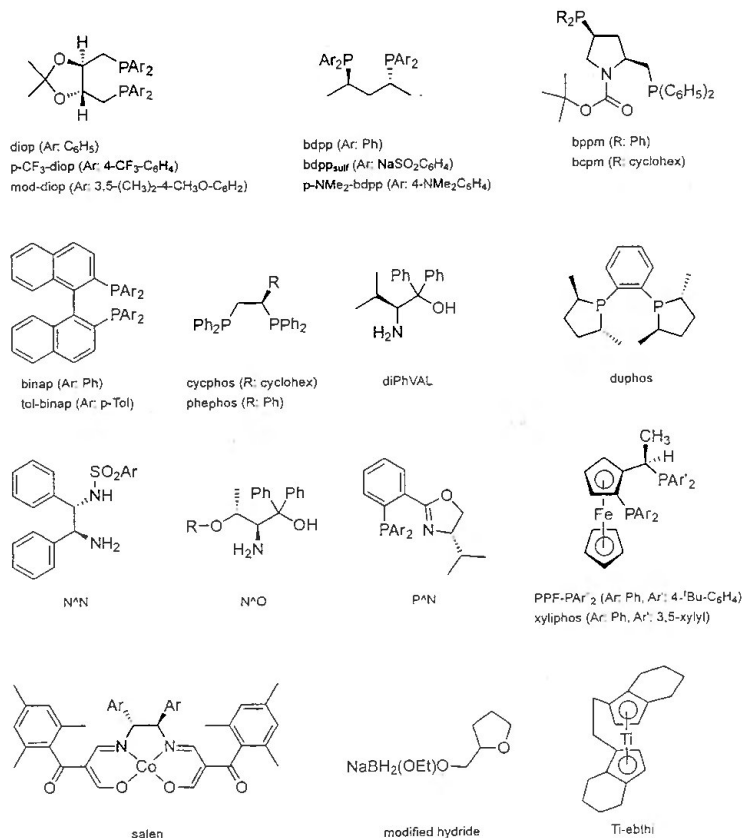


Fig. 1. Structures and abbreviations for chiral ligands

crease in activity and productivity. The reactions usually have to be carried out at relatively high hydrogen pressures. A production process was developed for the hydrogenation of the MEA imine, an intermediate for the Novartis herbicide Dual Magnum (Table 2, entry 2.2) using Ir-xyliphos in the presence of iodide and acid with 80% ee and extremely high s/c ratio and tof values [20].

For the model substrate *N*-phenyl-(1-phenylethylidene)amine, best results were obtained with an Ir phosphanodihydrooxazole catalyst (Table 2, entry 2.5) [23]. The highly enantioselective hydride reduction of the di-imine of entry 2.6 was accomplished with sodium borohydride and sub-stoichiometric amounts of a chiral amino alcohol as catalyst [24].

Table 2. Synthetically useful catalysts for the enantioselective reduction of *N*-arylimines **1** (see Fig. 2): Effective catalysts, reaction conditions, enantioselectivity, activity, and productivity

Entry	R	R'	R''	Catalyst	pH ₂ ^a (bar)	T (°C)	ee (%)	s/c	tof (h ⁻¹)	Ref
2.1			DMA-imine 1a	Ir-bdpp	40	30	90	500	40	22
2.2			MEA-imine 1b	Ir-xyliphos	80	50	80	1,000,000	>200,000	20
2.3			MEA imine 1b	Ir-PPF-PAr' ₂	70	-15	87 ^b	5000	30	39
2.4	Me	CH ₂ OMe	2,5-(Me) ₂ -thioph	Ir-PPF-PAr' ₂	20	5	80 ^{b)}	100	27	39
2.5	Me	Ph	Ph	Ir-P^N	100	5	89	1000	ca. 80	23
2.6		Ph(C=N-Ar)-(C=N-Ar)Ph		B-N^O	BH ₃	r.t.	99 ^c	200	<1	24

^a or reducing agent^b optimized for selectivity^c of the diamine**Table 3.** Effect of imine structure on catalyst performance for the enantioselective reduction of *N*-arylimines **1** with Ir-diop iodide at 20 bar H₂ and r.t. [17]

Entry	R	R'	R''	ee (%)	tof (h ⁻¹)
3.1	Me	CH ₂ OMe	2-Me-6-Et-Ph	62	200
3.2	Me	CH ₂ OMe	2,6-(Me) ₂ -Ph	68	3.6
3.3	Me	n-Pr	2,6-(Me) ₂ -Ph	52	3.6
3.4	Me	n-Pr	Ph	16	18.5
3.5	Me	CH ₂ OMe	2,5-(Me) ₂ -thioph	58	27
3.6	Me	Ph	Ph	22	5

The only systematic study of the effect of the imine structure on enantioselectivity was reported in connection with the development of the Ir system for the hydrogenation of the MEA imine (see Table 3) [17]. With Ir-diop iodide as catalyst, both the 2,6-alkyl substituents (either Me or Et) of the *N*-phenyl group as well as the methoxy-substituent of R' contribute to the high enantioselectivity. Replacing the methoxy group of the DMA-imine by an ethyl group led to a decrease in ee from 69% to 52%; further replacement of the 2,6-dimethylphenyl group by phenyl gave an ee of 16% (Table 3, entries 3.2, 3.3, and 3.4). It is noteworthy that the 2,6-dimethylphenyl group could be replaced by a 2,5-disubstituted thien-3-yl group without loss in catalyst activity (Table 3, entry 3.5). These results indicate that the presence of an *ortho*-disubstituted ring is the essential requirement for good enantioselectivity. This in contrast to the result obtained for the Ir-phosphanodihydrooxazole system [23].

3.2 Enantioselective Reduction of *N*-Alkylimines

Up to now, most studies with *N*-alkylimines (Fig. 2) were carried out with model compounds, especially with *N*-benzyl-(1-phenylethylidene)amine (and some analogs). One reason for this choice could be the easy preparation of a pure crystalline starting material, another that the primary amines are accessible by hydrogenolysis of the benzyl group. As can be seen in Table 4, Rh-bdpp_{sulf} gave the highest enantioselectivity ever reported for the hydrogenation of an acyclic imine [16]. Rh-cycphos is also able to hydrogenate *N*-alkyl-(1-phenylethylidene) amines with >90% ees, but with rather low catalyst activity [25]. The novel Ir-phosphanodihydrooxazole (Ir-P[^]N) catalyst (Table 4, entries 4.3 and 4.4) is probably not yet preparatively useful for this transformation (moderate ees, low s/c and tof) [23]. *N*-Alkylimines of cyclic ketones can be reduced with moderate to good enantioselectivities using a Ru-diamine complex and formate as transfer

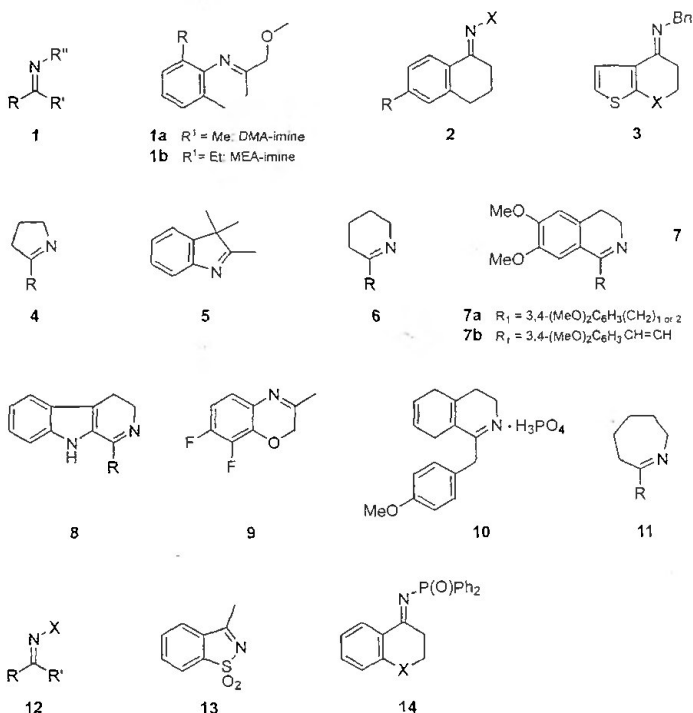


Fig. 2. Structures of substrate molecules

Table 4. Synthetically useful catalysts for the enantioselective reduction of various *N*-alkylimines: Catalyst, reaction conditions, enantioselectivity, activity, and productivity

Entry	C=N	R	R'	R''	Catalyst	pH ₂ ^a (bar)	T (°C)	ee (%)	s/c	tof (h ⁻¹)	Ref
4.1	1	Me	Ph	Bn	Rh-bdpp _{sulf}	70	20	94–96	100	16–85	16
4.2	1	Me	4-(MeO)-Ph	Bn	Rh-cycphos	100	-25	91	100	0.7	25
4.3	1	Me	p-Tol	Bn	Ir-P [^] N	100	r.t.	79	25	2	23
4.4	1	Me	Ph	n-Bu	Ir-P [^] N	100	r.t.	75	25	2	23
4.5	2	H		Bn ^b	Ru-N [^] N	F	28	89	100	15	19
4.6	3				Ru-N [^] N	F	28	85–88	100	7–17	19

^aor reducing agent, F = HCOOH-Et₃N^bX group

reducing agent (Table 4, entries 4.5 and 4.6) [19]. In general, low s/c ratios have been applied and catalytic activities are very low for all catalysts described in Table 4 except for Rh-bdpp_{sulf} (tof up to 85 h⁻¹).

3.3

Enantioselective Reduction of Cyclic Imines (C=N Group Endocyclic)

Besides the cyclic model compounds 4, 6, and 11, a number of endocyclic imines were investigated because they are intermediates for the synthesis of chiral compounds with biological activity. Indeed, there is a surprisingly high number of substrates that can be hydrogenated with preparatively useful selectivities as depicted in Table 5. Cyclic imines do not have the problem of *syn/anti* isomerism, therefore, in principle higher enantioselectivities can be expected. The most impressive difference in this respect was observed for Ti-ebthi-catalyzed hydrogenations that give ees up to 99% for several cyclic imines whereas non-cyclic imines are reduced with ees <85% [6]. Most hydrogenation reactions have to be carried out at relatively high hydrogen pressures and at low to moderate temperatures.

Much effort was dedicated to the reduction of dihydropapaverin or analogs thereof (7 in Fig. 2). The resulting chiral amines are intermediates for natural and unnatural opioids. Very good enantioselectivities were reported for Ti-ebthi (Table 5, entry 5.6; see also entries 5.1 and 5.13) but tofs are rather low and the catalyst is quite difficult to handle. The Ru-catalyzed transfer hydrogenation with triethylammonium formate (Table 5, entries 5.7 and 5.8) and Ir-bcpm-catalyzed hydrogenation [26] (Table 5, entry 5.9) are also successful for imines of type 7, both with moderate tofs. Similarly, the hydrogenation of 8 (Table 5, entry 5.10) leads to alkaloids with biological activity and again Ru-N[^]N is the catalyst of choice [19]. Compound 9 is an intermediate for a new antibacterial agent and can be hydrogenated by Ir-diop to give ees up to 85% (Table 5, entry 5.11) [27]. The formally diastereoselective reduction of folic acid in aqueous phase was accomplished with an Ir-bppm complex adsorbed on silica, whereas in homogeneous

Table 5. Synthetically useful catalysts for the enantioselective reduction of cyclic imines: Effective catalysts, reaction conditions, enantioselectivity, activity, and productivity

Entry	C=N	R	Catalyst	pH ₂ ^a (bar)	T (°C)	ee (%)	s/c	tof (h ⁻¹)	Ref
5.1	4	Alk, Ph	Ti-ebthi	5	45–65	99	100	0.6–2	6
5.2	5		Ir-bcpm	100	–30	91	200	2	30
5.3	5		Ir-Ar ₂ PF-PAr' ₂	70	15	94	250	14	39
5.4	6	Ph	Ti-ebthi	33	65	98	20	0.7	6
5.5	6	Ph	Ir-tol-binap	60	r.t.	91	100	5	29
5.6	7	H	Ti-ebthi	140	65	98	20	0.4	6
5.7	7	Ar	Ru-N [^] N	F	28	84	200	6–25	19
5.8	7a		Ru-N [^] N	F	28	92–95	200	15–30	19
5.9	7b		Ir-bcpm	100	2	86	200	6	26
5.10	8	Me, Ph	Ru-N [^] N	F	28	96–97	1000	35–83	19
5.11	9		Ir-diop	40	0	85	100	–	27
5.12	10		Ir-Ph ₂ PF-P(<i>t</i> -Bu) ₂	70	r.t.	89	1500	75	31
5.13	11	Ph	Ti-ebthi	33	45	98	20	0.7	6
5.14	folic acid		Rh-bppm ^b	50	80	92 ^c	40	<2	28

^aor reducing agent, F = HCOOH·Et₃N^badsorbed on SiO₂^cdiastereomeric excess of tetrahydrofolic acid

solution this complex showed very low selectivity (Table 5, entry 5.14) [28]. Ir-diphosphine catalysts allow the hydrogenation of model compounds 6 (Table 5, entry 5.5) and 5 (Table 5, entries 5.2 and 5.3) as well as of 10, an intermediate for dextromethorphan (Table 5, entry 5.12) in 89–94% ee with low to good activity and productivity. A technical process has been developed for the latter reaction [31].

3.4

Enantioselective Reduction of Miscellaneous C=N-X Systems

Oximes that are easy to prepare and readily available were among the first C=N derivatives to be tried as starting material for the catalytic preparation of chiral amines. However, with the exception of entry 6.1 in Table 6, all results were disappointing especially for oximes of α -keto acid derivatives leading to α -amino acids [27]. In contrast, quite effective catalysts have been developed that can reduce related C=N-X compounds (suitable are NHCOR, P(O)Ph₂, and Ts) with good to very high ees and in some cases with good catalyst activities as well. A disadvantage of these types of substrate is the fact that the X group has to be split off in order to obtain the corresponding amine, whereas it is usually produced directly when reducing an oxime. Preparatively most useful is probably the hydrogenation of *N*-acyl hydrazones with the Rh-duphos catalyst (Table 6, entries 6.3 and 6.4). This reaction was developed by Burk and Feaster [18] in analogy to

Table 6. Synthetically useful catalysts for the enantioselective reduction of miscellaneous C=N-X systems: Effective catalysts, reaction conditions, enantioselectivity, activity, and productivity

Entry	C=N	R	R'	X	Catalyst	pH ₂ ^{a)} (bar)	T (°C)	ee (%)	s/c	tof (h ⁻¹)	Ref
6.1	12	Me	Ph	OMe	diPhVAL	BH ₃	r.t.	90	4	0.2	32
6.2	12	Me	Ph	P(O)Ph ₂	Co-salen	"H"	0	90	100	34	33
6.3	12	Me	Ar	NHCOPh	Rh-duphos	4	0	88–96	500	14–42	18
6.4	12	Ph	COOMe	NHCOPh	Rh-duphos	4	20	91	500	1000	18
6.5	12	Ph	Et	Ts	Ru-binap	70	40	84	20	<1	33
6.6	2	H		Ts	Ru-binap	70	40	82	20	<1	33
6.7	13				Ru-binap	4	r.t.	99	90	6	34
6.8	2	H, OMe		P(O)Ph ₂	Co-salen	"H"	0	98–99	100	22–25	35
6.9	14				Co-salen	"H"	0	91–94	100	20	35

^aor reducing agent ("H", modified hydride see Fig. 1)^bX = O, CH₂CH₂ or direct bond

the well known hydrogenation of *N*-acyldehydroamino acid derivatives. The activity of the system is quite high with tofs up to 1000 h⁻¹. Several *N*-sulfonylimines can be hydrogenated with Ru-binap in high to very high optical yields but low activities (Table 6, entries 6.5 to 6.7). Finally, several types of *N*-diphenylphosphinylimines are reduced with a modified borohydride in presence of Co-salen complexes (Table 6, entries 6.2, 6.8, and 6.9). The enantioselectivities are very impressive but activities and s/c ratio are relatively low [35].

4

Discussion of Effective Catalysts

4.1

Homogeneous Metal Catalysts

All effective catalysts for the asymmetric reduction of prochiral C=N groups are based on complexes of rhodium, iridium, ruthenium, and titanium. Whereas in early investigations (before 1984) emphasis was on Rh and Ru catalysts, most recent efforts were devoted to Ir and Ti catalysts. In contrast to the noble metal catalysts which are classical coordination complexes, Buchwald's *ansa*-titanocene catalyst for the enantioselective hydrogenation of ketimines represents a new type of hydrogenation catalyst [6]. In this chapter important results and characteristics of effective enantioselective catalysts are summarized.

4.1.1

Rhodium-Diphosphine Complexes

Most rhodium catalysts for the enantioselective reduction of the C=N group are prepared in situ from a dimeric Rh-diene complex and a chiral diphosphine. Only few of the tested diphosphine ligands exhibit enantioselectivities >70%: bdpp, cycphos, and phephos for imines and duphos for acylhydrazones. The activity of most Rh-diphosphine complexes for imine hydrogenation is low and therefore most of them are of limited practical use. Although some catalysts work already at ambient reaction conditions, most Rh-diphosphine complexes show low *tofs* even at elevated hydrogen pressures (>60 bar).

Interestingly, it was found that the highest *ees* were obtained with sulfonated diphosphines (bdpp_{sulf}) in an aqueous bi-phasic medium [16]. The degree of sulfonation affected the optical yield. For the hydrogenation of *N*-benzyl-*N*-(1-methylbenzylidene) amine in ethyl acetate-H₂O as solvent, the mono-sulfonated bdpp gave 94% *ee*, compared to 65% *ee* with Rh-bdpp in MeOH. In addition, the activity of the mono-sulfonated catalyst was higher by a factor of 5. A similar increase in enantioselectivity was reported for [Rh(nbd)(bdpp)]ClO₄ as catalyst precursor in presence of reverse AOT micelles (*ee* 92% vs. 80% in neat benzene) [36]. However, it appears that in this case the sulfonate anion, and not the micellar structure, is responsible for the observed increase in optical yields since addition of non-surfactant sulfonate salts also led to higher enantioselectivity. Different catalytic properties are also observed for neutral diphosphine catalysts prepared from [Rh(nbd)Cl]₂ and the corresponding cationic catalysts [Rh(nbd)diphosphine]⁺. It seems that the presence of halide ions affects both the hydrogenation rate as well as the enantioselectivity, albeit differently for different substrates. Whereas the asymmetric hydrogenation of *N*-benzylketimines is affected favorably by the addition of halides [15, 25], the Rh-catalyzed hydrogenation of DMA-imine is inhibited by the presence of iodide [25]. In some cases, the neutral and the cationic catalysts even lead to the opposite product enantiomer [15]. Obviously, the coordination sphere of the rate- and product-determining species is quite different for the two types of Rh complexes.

The reproducibility of rate and enantioselectivity can be a problem as described by the Marko group for the cationic catalyst [Rh(nbd)diphosphine]⁺ and *N*-benzyl-*N*-(1-methylbenzylidene)amine as model substrate [15]. Extensive efforts to find a correlation between purity of imine, water and oxygen content and the catalyst performance were not successful. However, these problems are not uncommon for enantioselective catalytic reactions but were not addressed any more in reports published at a later stage by Marko or other groups.

4.1.2

Iridium Complexes

In the last decade, several types of Ir-based catalysts were developed and applied successfully. The most efficient type, prepared from [Ir(cod)Cl]₂, a chiral di-

phosphine and a halide (iodide), proved to be much more active for the enantioselective hydrogenation of *N*-arylketimines than the rhodium analogs [17]. Despite a significant tendency for deactivation (strongly dependent on temperature, ligand structure, solvent and the basicity of the formed aniline derivative) several Ir catalyst are of practical use for the production of enantiomerically enriched *N*-arylamines. Under optimized reaction conditions, particularly highly pure starting materials, turnover numbers of >1000 can be obtained with a high reaction rate. Additives play a dominant but unpredictable role, acting either as promoters (acid, iodide, amines) or as deactivators (amines, iodide). Enantioselectivities in the range of 80% to >90% were achieved for several types of imines with the newly developed ferrocenyldiphosphines [37] and bdpp; diop also sometimes gives useful results.

For the commercially important MEA hydrogenation the simultaneous presence of acid and iodide promoters was reported to be a critical factor (besides the ligand structure of course) [20]. The dramatic effect was illustrated with a series of screening experiments using a catalyst generated *in-situ* from $[\text{Ir}(\text{cod})\text{Cl}]_2$ and xyliphos. Without or with only one of these additives, the time for complete conversion was in the range of 10–16 h (s/c 800). When both tetrabutylammonium iodide and acetic acid were used, the initial rates increased by a factor of 5 and the time for complete conversion decreased by a factor of 20. The amount of acid necessary for full effect depends on the acid strength. Very strong acids such as H_2SO_4 can be used in extremely small concentrations, as long as the ratio of acid/Ir is >20. Weak acids are effective only at higher concentrations.

Systematic structure-activity relationship studies were carried out with Ir-diphosphine catalysts generated from $[\text{Ir}(\text{cod})\text{Cl}]_2$ and iodide using MEA imine as substrate [20]. Some representative results are summarized in Table 7. Both chelate size and the nature of the substituent at the phosphorus atom affected the catalytic properties of the Ir catalysts. The highest activities as well as good optical yields were obtained with 1,3- and 1,4 diphosphines, in particular with bdpp, diop, and ferrocenyldiphosphines of the type $\text{Ar}_2\text{PF-Par}'_2$. A comparison of the *tof* after 4 h and at the end of the reactions shows that Ir-ferrocenyldiphosphines catalysts are more stable than with classical ligands (Table 7, entries 7.1 to 7.6). The ligand bppm gave the best enantioselectivities but showed low activity even at 80 bar (Table 7, entry 7.1). Diop and substituted diop ligands showed similar catalyst performances (Table 7, entries 7.2 to 7.4). However, exchanging the Ph group in bdpp with *p*-(Me)N-Ph led to racemic product (Table 7, entries 7.5 and 7.6). 1,2-Diphosphines such as dipamp, norphos, or chiraphos but also bppfoh were rather unselective (results not shown [20]). Under the test conditions, Ir catalysts containing PPF- Par'_2 (Table 7, entries 7.8 to 7.14) had quite a high inherent activity and showed little tendency towards deactivation, whereas basic phosphines such as PPF-Pcy₂ (Table 7, entry 7.7) were not very active. Especially PPF-P(3,5-xyl)₂ (xyliphos) turned out to give an exceptionally active catalyst and, even more importantly, it did not deactivate! With this catalysts s/c >10,000 and complete conversion could be achieved. Even from the limited number of examples in Table 7 it is clear that both steric and electronic proper-

Table 7. Hydrogenation of DMA imine with $[\text{Ir}(\text{cod})\text{Cl}]_2$ -iodide. Effect of diphosphine structure [20]

Entry	ligand	solv.	T °C	pH ₂ bar	tof ₅ ^a [h ⁻¹]	ee [%]
7.1	bppm	B	10	80	6	79
7.2	diop	A	25	25	32 (165)	61
7.3	mod-diop	A	25	25	28 (146)	59
7.4	<i>p</i> -CF ₃ -diop	A	25	25	32 (89)	61
7.5	bdpp	A	-5	25	26 (114)	78
7.6	<i>p</i> -NMe ₂ -bdpp	C	25	25	31	rac
7.7	PPF-Pcy ₂	C	25	50	4	n.d.
7.8	PPF-P(<i>o</i> -MeO-Ph) ₂	C	25	30	132 (190)	7
7.9	PPF-P(<i>m</i> -MeO-Ph) ₂	-	25	25	43 (82)	64
7.10	PPF-P(<i>p</i> -MeO-Ph) ₂	-	25	25	44 (120)	62
7.11	PPF-P(<i>m</i> -tol) ₂	-	25	25	45 (86)	70
7.12	PPF-P(<i>p</i> -CF ₃ -Ph) ₂	C	25	30	133 (170)	56
7.13	PPF-P(3,5-xyl) ₂	C	80	50	331 (950)	75
7.14	PPF-P(3,5-xyl) ₂	-	80	50	807 (1450)	69

Reaction conditions: $[\text{Ir}(\text{cod})\text{Cl}]_2$, *n*-Bu₄I, solvent: A: MeOH/tol (1:1); B: *t*-BuOCH₃; C: THF/CH₂Cl₂ (3:1), s/c: entries 1–6: 200, entries 7–13: 800, entry 14: 10,000.

^a in parenthesis: tof after 4 h

ties of the Ar' group affect the activity as well as the enantioselectivity of the Ir-PPF-PAr'₂ catalysts.

Strong iodide effects were also observed for the hydrogenation of *N*-benzyl-*N*-(1-methylbenzylidene)amine and of the cyclic imine **5** with $[\text{Ir}(\text{cod})\text{Cl}]_2$ and diop and bppm (and analogs thereof) as ligands. In the presence of *n*-Bu₄NI and BiI₃ as promoters and an s/c ratio of 200, complete conversion were achieved for both substrates with ees up to 91% for imine **5** [30].

An interesting effect of imide and amine additives was observed for Ir-bcpm and Ir-binap catalysts. The catalyst performance for the hydrogenation of imines **7** was affected by the addition of phthalimide or perfluorophthalimide [26]. The highest enantioselectivity were obtained with the Ir-bcpm catalyst system: 87% ee with F₄-phthalimide and 86% with phthalimide. Primary or secondary amines were found to be useful co-catalysts for the asymmetric hydrogenation of *N*-benzyl-*N*-(1-methylbenzylidene)amine and the cyclic imine **6** (R=Ph) with cationic Ir-binap (or Ir-tol-binap) catalysts [27]. For example, the addition of 5 equiv./Ir-benzylamine increased the ee values from 20% to 70% with *N*-benzyl-*N*-(1-methylbenzylidene)amine and from 40% to 90% with **6** (R=Ph), respectively. Surprisingly, the addition of benzylamine to a neutral in situ Ir/tol-binap complex led to complete catalyst inhibition in the hydrogenation of imine **6** (R=Ph), whereas in absence of this additive complete conversion as achieved with up 89% ee. The role of these co-catalysts is not well understood.

Halogen-free catalysts of the type $[\text{Ir}(\text{diphosphine})(\text{OOCR})_3]$ and $[\text{Ir}(\text{diphosphine})(\text{OOCR})_2\text{H}]$ ($\text{R}=\text{CH}_3$, CF_3), obtained by treating $[\text{Ir}(\text{diphosphine})\text{HI}_2]_2$ with $\text{Ag}(\text{OOC}\text{CF}_3)$ [22], were also active for the hydrogenation of *N*-arylimines but showed a different catalytic profile. For example, with $[\text{Ir}(\text{bdpp})(\text{OOC}\text{CF}_3)_3]$ as catalyst precursor, DMA imine could be hydrogenated with 90% ee at 0°C , while the cyclic imine **5** was reduced with only a moderate ee of 35%. It was suggested that the catalysis proceeds in part via initial dissociation of the carboxylate ligand.

Two novel ligand types were recently published that form effective Ir complexes. Osborn and Sablong described a C_2 -symmetric tridentate PNP-ligand that enables the hydrogenation of DMA imine with up to 55% ee [38]. Pfaltz and coworkers [23] reported the development of Ir-phosphanodihydrooxazole ($\text{P}^\wedge\text{N}$) catalysts for the hydrogenation of both *N*-aryl- and *N*-alkylimines. Using cationic Ir complexes of the type $[\text{Ir}(\text{cod})\text{P}^\wedge\text{N}]\text{PF}_6$, *N*-arylimines were hydrogenated with ees up to 89%, *N*-alkylimines with a maximum of 79% ee. In this case, the addition of iodide resulted in a dramatic decrease of the enantioselectivity and a change of the absolute configuration of the major enantiomer.

4.1.3

Ansa-Metallocene-Titanium Complexes

A very different type of catalyst was developed by Buchwald et al. [6]: the chiral Ti complex with Brintzinger's ansa-metallocene ligand (ebthi) is extraordinarily effective for the enantioselective hydrogenation of cyclic imines with high optical yields (>97% ee). Unfortunately, the activity and productivity of this Ti catalyst are relatively low compared to Rh- and Ir-diphosphine catalysts. The stereochemical outcome of the reaction can be predicted by straightforward steric arguments. Acyclic imines are reduced with lower enantioselectivity, probably due to isomerization problems and the presence of *syn/anti* isomers. Studies with multifunctional imines or in presence of other substrates revealed that the scope of the Ti-ebthi catalyst is rather limited. Partial or total catalyst inhibition is observed in presence of most functional groups, except amines, alcohols, acetals, and halides [39].

4.1.4

Ruthenium Complexes

In general, Ru-binap catalysts which show excellent enantioselectivity in the asymmetric reduction of functionalized olefins and ketones are not very suitable for $\text{C}=\text{N}$ hydrogenations. An exception is the hydrogenation of an *N*-sulfonylimine: 99% ee for **13** [34] and 82% and 84% for **2** and **12**, respectively (see Table 6). Other Ru-diphosphine complexes were investigated but proved to be little active and selective [2]. James and coworkers [40] investigated the catalytic activity of several mono and binuclear Ru-containing chiral diphosphines such as $\text{RuCl}_2(\text{diphosphine})(\text{RCN})_2$, $\text{RuCl}_2(\text{diphosphine})(\text{diene})$, $\text{Ru}_2\text{Cl}_4(\text{diphosphine})_2(\text{NEt}_3)$,

or $\text{Ru}_2\text{Cl}_5(\text{diphosphine})_2$ for the enantioselective hydrogenation of imines of type 1. However, all these catalysts showed both low activity and enantioselectivity. The highest optical yield was obtained with $\text{Ru}_2\text{Cl}_5(\text{chiraphos})_2$ (27% ee).

A major breakthrough was achieved with transfer hydrogenation catalysts generated from $[\text{Ru}(\text{Cl}_2(\text{arene}))]$ and a semi-*N*-sulfonated chiral diamine (e.g., *N*-tosyl-1,2-diphenyl-1,2-diaminoethane, $\text{N}^{\wedge}\text{N}$) [19]. With $\text{HCOOH}/\text{NEt}_3$ as donor, enantioselectivities up to 97% were obtained for substrates of types 7 and 8 (s/c: 200–1000; tof: 83 h^{-1}), providing a new general route to alkaloids (Tables 4 and 5). Furthermore, this catalyst exhibited a high chemoselectivity towards the reduction of $\text{C}=\text{N}$ vs. $\text{C}=\text{O}$ functions. As a consequence, imines of type 7 can even be reduced in acetone with <5% production of 2-propanol.

4.2

Modified Hydrides and Heterogeneous Catalysts

Hydride reductions of $\text{C}=\text{N}$ groups are well known in organic chemistry. It was therefore obvious to try to use chiral auxiliaries in order to render the reducing agent enantioselective [41]. The chiral reagent or catalyst is prepared by addition of a chiral diol or amino alcohol and the active species is formed by reaction of OH or NH groups of the chiral auxiliary with the metal hydride. A major drawback of most hydride reduction methods is the fact that stoichiometric or higher amounts of chiral material are needed. At this time only two such catalytic systems are useful for preparative purposes (Table 2, entry 2.6, Table 6, entry 6.1) [24, 32].

The most important heterogeneous systems for the hydrogenation of $\text{C}=\text{N}$ groups have been reviewed by Blaser and Müller [8]. Neither the use of soluble modifiers nor of chiral supports led to heterogeneous catalysts with useful enantioselectivities. The interaction between adsorbed substrate, the active site and the chiral auxiliary employed until now was probably not sufficient for a good discrimination.

4.3

Mechanistic Aspects

Only a few detailed studies of the reaction mechanism of the homogeneous hydrogenation of imines have been published up to now. A generalization seems to be very difficult for two reasons. First, rather different catalyst types are effective and probably act by different mechanisms. Second, the effect of certain additives (especially iodide and acid/base) is often decisive for ee and rate but a promoter in one case can be a deactivator in another one.

For Rh- and Ir-diphosphine-based catalysts there exist some indications on reactive species and also on hydrogen activation. James and coworkers [2, 25] investigated the Rh-catalyzed DMA-imine hydrogenation and concluded that the imine is η^1 -coordinated via the nitrogen lone pair to the Rh center and not via the π -system of the $\text{C}=\text{N}$ bond. They also suggested that the hydrogen activation

occurs after the imine is coordinated. Osborn and Chan [42] investigated, isolated, and characterized Ir(III) complexes of the type: $[\text{Ir}(\text{diphosphine})\text{I}_4]^-$, $[\text{Ir}(\text{diphosphine})\text{HI}_2]_2$, and $[\text{Ir}(\text{diphosphine})\text{I}_3]_2$. All three Ir complexes were found to be catalytically active for the hydrogenation of DMA-imine, suggesting the formation of the same active monomeric Ir species as for the in situ formed catalyst by splitting the iodo-bridge.

Based on these results, the catalytic cycle depicted in Fig. 3 can be postulated: Starting species is an Ir(III)-H species that coordinates the imine via the lone pair in a η^1 -manner (as proposed for the Rh-catalyzed reaction). A η^1, η^2 -migration leads to two diastereomeric adducts with a π -coordinated imine that then inserts into the Ir-H bond to give the corresponding Ir-amide complexes. The last step is a simultaneous hydrogenolysis of the Ir-N and the formation of an Ir-H bond, presumably via heterolytic splitting of the dihydrogen bond. In contrast to the Rh-diphosphine catalyzed hydrogenation of C=C bonds that most likely occurs via Rh(I) and Rh(III) species, the cycle in Fig. 3 consists exclusively of Ir(III) species. It is obvious that this basic catalytic cycles neither explains the mode of enantioselection nor the sometimes dramatic effects of additives, e.g., the strong rate enhancement by acids observed for the Ir-xyliphos-catalyzed MEA-imine hydrogenation.

A similar mechanism was proposed for the Ti-catalyzed reactions by Buchwald [6]. The active catalyst, that is produced by reacting ebthi-TiR₂ with *n*-BuLi followed by phenylsilane, was proposed to be the monohydride species ebthi-TiH. Kinetic and deuterium labeling studies are in agreement with the following reaction sequence: ebthi-Ti-H reacts with the imine via 1,2-insertion reaction to form two diastereomeric Ti-amide complexes. Then this intermediate amide complexes reacts via a σ bond metathesis reaction with dihydrogen to regenerate the titanium hydride and to form the two amine enantiomers. The reaction of the titanium amide complex with molecular hydrogen is proposed to be the

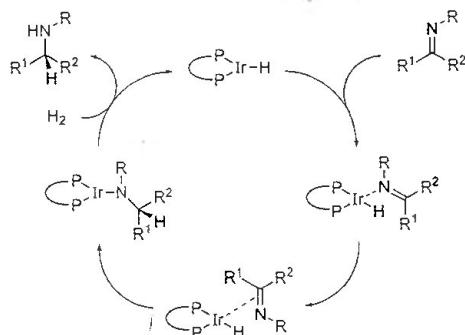


Fig. 3. Schematic catalytic cycle postulated for the Ir-diphosphine catalyzed hydrogenation of N-arylimines. Not shown are halide ligands

rate-determining step. The discrimination of the catalyst is due only to the size difference of the imine substituents, thereby explaining the potential detrimental effect of the presence of *syn* and *anti* isomers. The absolute configuration of the major enantiomer can be predicted by simple sterical arguments assuming that the 1,2-insertion is the product determining step.

5

Assessment of Methods

Different criteria are important for assessing the applicability of a catalyst either for preparative purposes or the technical manufacture of chiral fine and specialty chemicals [4, 44]. In both cases, enantioselectivity is of course the decisive prerequisite. For preparative use, easy availability and handling of the catalyst will probably play a major role. For technical applications, catalyst activity (tof), productivity (s/c ratio), availability on a large scale, and of course cost also play an important role. In general, optical yields should be >90%, unless a further enrichment is easy. The minimal activity and productivity required is less predictable. As an example: for an s/c ratio of 1000 and a tof of 10 h^{-1} the reaction will take 100 h for completion, probably not an acceptable reaction time for a large volume chemical; for preparative use, s/c could be decreased to 100 and the reaction time would be an acceptable 10 h. Table 8 gives a summary on the range of reaction conditions, ees, s/c ratios, and tofs for important chiral catalysts and substrates. An overall assessment for the different systems takes the catalytic properties of a catalyst for a given transformation but also ecological and economical aspects into consideration.

Table 8. Typical ranges of reaction conditions, optical yields, turnover frequencies, and substrate to catalyst ratios for various reactions with preparative potential

Catalyst	Substrate type	pH ₂ (bar)	T (°C)	ee (%)	s/c	tof (h ⁻¹)
Ir diphosphine	N-arylimines	20–80	0–30	80–90	1000–1'000'000	6–>200,000
	cyclic imines	20–40	20–30	85–94	100–1500	2–75
Ir-P^N	imines	100	r.t.	75–89	25–1000	1–>1000
Rh diphosphine	imines	60–100	<0–30	91–96	40–1000	0.1–85
	acyl-hydrazones	4	<0–20	70–96	500–1000	10–1000
Ru diphosphine	C=N-SO ₂ R	4–70	r.t.–40	82–99	20–90	1–6
Ru-N^N	imines	formate	30	85–97	100–1000	6–83
Co-N^N	C=N-P(O)Ph ₂	hydride	0	88–99	100	20–34
Ti-ebthi	cyclic imines	5–33	20–65	98–99	20–100	0.4–2.4

5.1

Catalytic Methods

5.1.1

Enantioselectivity

The results summarized in Table 8 demonstrate that the catalytic reduction of various C=N functions has reached the state of being preparatively useful for a variety of amines. *Catalyst productivity, costs:* Most metal complexes and all ligands are very expensive. If the sometimes rather low s/c ratios used in most studies are not improved, catalyst costs would be medium to high. *Ecological aspects:* Hydrogenation reactions have a very good ecological balance since no stoichiometric by-products are produced. Hydrides and modified formates as reducing agents usually give stoichiometric amounts of by-products that must be disposed of.

5.1.2

Overall Assessment

At the moment, the production process made possible by the extraordinarily active and productive Ir-xyliphos catalyst is the exception, but there are indications that other technical applications might be feasible. For preparative applications in the laboratory some „catalytic know-how“ is required to make these catalysts work. The scope of the various catalyst types is sometimes still rather narrow, but will most likely improve with time. However, in most cases activity and productivity must be improved decisively. As more and more types of ligands are becoming commercially available a broader catalyst screening will become possible. This means that progress regarding both enantioselectivity and activity is quite likely.

5.2

Alternative Reduction Methods

Several *stoichiometric chiral reducing agents* starting from BH_3 , LiAlH_4 or NaBH_4 and relatively cheap amino alcohols or diols have been developed for the reduction of imines and oxime derivatives [3, 43]. Yields are medium to very high. Most effective chiral auxiliaries can be prepared in one or two steps from rather cheap starting materials like amino acids, tartaric acid, or sugars and they can probably be recycled. Because these reagents are used in stoichiometric amounts, the costs for the reduction step will be medium to high anyway. *Overall assessment:* Chiral hydrides are at the moment useful on a laboratory scale but their potential for technical applications is medium to low.

Hydrosilylation seems at the moment not very attractive for C=N functions [3] (also see chap. 6.3). With one exception [45], yields are low to medium and catalyst activity and productivity are low to medium.

Chiral amines can also be produced using *aminotransferases* either by kinetic resolution of the racemic amine or by asymmetric synthesis from the corresponding prochiral ketone. The reaction involves the transfer of an amino group, a proton, and two electrons from a primary amine to a ketone and proceeds via an intermediate imine adduct. A variety of chiral amines can be obtained with high to very high yields. Several transformations have been developed and can be carried out on a 100 kg scale [46].

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